

When transistors are used in digital circuits they usually operate in the:

Select one:

- ☒ a. saturation and cutoff regions
- ☐ b. linear region
- ☐ c. active region
- ☐ d. breakdown region



If a 169.7 V half-wave peak has an average voltage of 54 V, what is the average of two full-wave peaks?

Select one:

- ☐ a. 119.9 V
- ☒ b. 108.0 V
- ☐ c. 115.7 V
- ☐ d. 339.4 V



Is a type of clipper which removes the negative half cycles of the input AC signal.

Select one:

- ☒ a. series negative clipper
- ☐ b. series positive clipper
- ☐ c. shunt positive clipper
- ☐ d. dual clipper



For a clamping circuit , Sometimes an independent dc supply is also required to cause an:

Select one:

- ☐ a. additional V peak output ,
- ☐ b. additional shift.
- ☐ c. additional frequency.
- ☒ d. additional V peak Input .



When the regulation by a Zener diode is with a varying input voltage, what happens to the voltage drop across the resistance?

Select one:

- ☒ a. Increases
- ☐ b. Has no effect on voltage
- ☐ c. The variations depend on temperature
- ☐ d. Decreases



The most frequent application for a _____ is in regulator networks and as a reference voltage.

Select one:

- ☒ a. Zener diode
- ☐ b. half-wave rectifier
- ☐ c. full-wave rectifier
- ☐ d. Ideal diode



In a certain biased limiter, the bias voltage is 5 V, and the input is 10 V peak sine wave. If the positive terminal of the bias voltage is connected to the cathode of the diode, the maximum voltage of the anode is:

Select one:

- ☒ a. 10 V
- ☐ b. 5.7 V
- ☐ c. 5 V
- ☐ d. 0.7 V



Is a type of clipper which removes the positive half cycles of the input AC signal.

Select one:

- ☐ a. shunt negative clipper
- ☐ b. series negative clipper
- ☒ c. series positive clipper
- ☐ d. dual clipper



Diode type that is used to regulate the output voltage is

Select one:

- ☐ a. Silicon Diode
- ☒ b. Zener
- ☐ c. LED
- ☐ d. Germanium

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To measure a current passes through an element, the Ammeter is placed in:

Select one:

- ☐ a. parallel
- ☐ b. No effect
- ☒ c. series
- ☐ d. (B and C)



A filtered full-wave rectifier voltage has a smaller ripple than does a half-wave rectifier voltage for the same load resistance and capacitor values. This statement is

Select one:

- ☐ a. False
- ☒ b. True



Diode clamper is:

Select one:

- ☒ a. Change the voltage level of an dc signal by adding a ac voltage
- ☐ b. Change the voltage level of an ac signal by adding a ac voltage
- ☐ c. Change the voltage level of an dc signal by adding a dc voltage
- ☐ d. Change the voltage level of an ac signal by adding a dc voltage

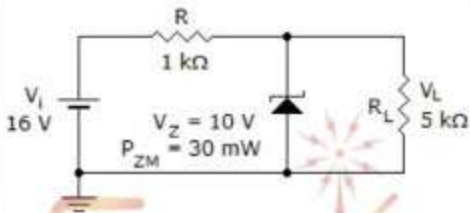
To get a peak load voltage of 40 Vout of a bridge amplifier. What is the approximate rms value of secondary voltage?

Select one:

- ☐ a. 0 V
- ☐ b. 14.4 V
- ☒ c. 28.3 V
- ☐ d. 56.6 V



Calculate I_L and I_Z



Select one:

- ☐ a. 2 mA, 0 mA
- ☐ b. 4 mA, 2 mA
- ☒ c. 4 mA, 2 mA
- ☐ d. 2 mA, 4 mA

A zener diode can be considered as :

Select one:

- ☐ a. Is forward biased
- ☐ b. Has a barrier potential of 1v
- ☐ c. A battery
- ☒ d. Has a constant voltage in the breakdown region

Depending on the type of the applied voltage, its resistance at any point on the curve can be determined

Select one:

- ☒ a. base resistance
- ☐ b. drain resistance
- ☐ c. static resistance
- ☐ d. dynamic resistance



clammer circuit is:

Select one:

- ☒ a. Usually Capacitance and resistance combination at the input
- ☐ b. Usually diode and resistance parallel combination at the output
- ☐ c. Usually Capacitance and resistance combination in parallel
- ☐ d. Usually Capacitance and diode parallel combination at the Input

When in its "on" state, the voltage across an ideal Zener diode, _____.

Select one:

- ☐ a. gets larger with an increase in applied voltage
- ☐ b. increases sharply with a decrease in applied voltage
- ☒ c. gets smaller with an increase in applied voltage
- ☐ d. None of these

When measuring an input Ac voltage using the DMM, the reading is given :

Select one:

- ☐ a. Peak Value
- ☐ b. Peak to Peak value
- ☐ c. RMS value
- ☒ d. All of Above

